Confidence-Guided Bayesian Network Parameterization with Language Models and Ranked Nodes: An Exploratory Study Across Two Domains

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Abstract

Large language models can propose conditional probability tables (CPTs) for Bayesian networks, but numerical plausibility and reliable parameterization are different requirements. We investigate a hybrid procedure that uses self-reported knowledge confidence to select rows for direct distribution elicitation and completes other rows with a Ranked Nodes approximation. We reconstruct the executed procedure from notebooks, raw responses, and evaluation archives for seismic risk and electric-vehicle fire risk. Two Qwen3.5 model sizes, a no-persona condition, and three domain personas yield 16 generated networks; equal and configured-weight persona pools yield eight additional outputs. We independently reproduce their archived probabilities and five agreement metrics, and add a uniform baseline. Results depend on domain, metric, and prompting condition. The 4B equal pool has the lowest weighted BN KL in both domains, yet all twelve electric-vehicle outputs have worse mean CPT KL than uniform probabilities. The 9B model selects more conditional rows for direct elicitation, including highly confident distributions with large reference disagreement. These single-run findings do not establish causal benefits of routing or statistical superiority. They identify a practical evaluation requirement for hybrid probabilistic modeling: audit routing decisions, extreme probabilities, simple baselines, and both local and reference-weighted errors before treating generated CPTs as usable knowledge.

CCS Concepts

• **Computing methodologies** → **Bayesian network models**; **Natural language generation**; *Knowledge representation and reasoning*.

Keywords

Bayesian networks, conditional probability tables, large language models, ranked nodes, confidence, probabilistic knowledge elicitation

1 Introduction

Bayesian networks (BNs) represent uncertainty through a directed acyclic graph and a conditional probability table (CPT) for each variable. A graph can be understandable while its parameters remain difficult to obtain: the number of conditional distributions grows with the combinations of parent states. When representative observations are unavailable, parameterization often depends on expert judgment and simplifying assumptions. Ranked Nodes methods address part of this burden by constructing distributions for ordered variables from a smaller set of parameters and qualitative relationships [2, 4].

Large language models (LLMs) offer another source of candidate probabilities. They can interpret variable descriptions, adopt domain roles, and generate numerical distributions. Recent work directly examines extracting probabilistic knowledge from LLMs for BN parameterization [6]. This capability raises an operational question: when should a model supply a complete distribution, and when should a constrained representation complete the CPT? An apparently confident answer may still disagree strongly with a benchmark. Conversely, a smooth, valid CPT generated by a compact mathematical model can be inappropriate for a particular mechanism.

We examine a confidence-guided hybrid procedure in two applied domains. It requests direct distributions for conditional rows whose self-reported knowledge confidence reaches a threshold, retains these rows as anchors, and completes the remaining rows using a midpoint weighted-mean Ranked Nodes approximation. Domain personas supply alternative elicitation contexts, whose outputs can also be pooled. Our focus is the behavior of this complete procedure under the recorded settings, including how routing and aggregation relate to observed errors.

The evidence consists of completed experiment artifacts rather than a supplied prospective protocol. We therefore reconstruct the executed design and distinguish original operations from retrospective analysis. The available corpus contains two reference networks, two model sizes, and four generation conditions, with one recorded run per cell. It cannot support claims about repeated-run variability or isolate the causal contribution of each component. This constraint is central to the interpretation.

We ask three questions. **RQ1:** How does reference agreement vary across domains, model sizes, and prompting conditions, including comparison with uniform probabilities? **RQ2:** How do equal and configured-weight persona pools compare with individual personas and the no-persona control? **RQ3:** Which rows are routed to direct elicitation, and what do the resulting error concentrations reveal about the confidence rule?

Our contributions are an auditable reconstruction of a hybrid LLM–Ranked Nodes experiment; an independently executable reanalysis of 16 generated and eight pooled networks, with simple baselines and explicit metric semantics; and a descriptive diagnosis of domain-dependent outcomes and failures among confidently elicited rows. The contribution is an exploratory assessment of a machine-learning application, not a claim that the hybrid method outperforms direct elicitation or expert parameterization.

2 Background and Related Work

For variables $X_1, \ldots, X_n$ and parent sets Pa_i, a BN represents $P(\mathbf{x}) = \prod_i P(x_i \mid \mathbf{x}_{\mathrm{Pa}_i})$. With a fixed graph and discrete states, parameterization specifies one probability vector per parent configuration, including one unconditional vector for each root. Normalization and non-negativity are necessary but do not establish correspondence to the intended domain.

Ranked Nodes encode ordered qualitative states through intervals on a latent scale. Fenton et al. [2] introduce a compact representation for qualitative judgments; Laitila and Virtanen [4] analyze its theoretical principle and applicability. Our implementation uses a weighted mean of parent-state midpoints and a truncated normal child distribution. It does not evaluate all Ranked Nodes aggregation families or integrate over uncertain parent intervals. We use "RN" below for this particular approximation, reserving broader conclusions about Ranked Nodes methods for future work.

Nafar et al. [6] study extracting numerical probabilistic knowledge from LLMs for BN parameterization, including elicitation of distributions and evaluation against existing networks. Their work motivates direct distribution elicitation as a relevant component and benchmark agreement as an informative, limited outcome. We examine a narrower hybrid setting in which model-reported confidence determines which rows receive direct elicitation and which receive constrained completion. Our two-domain experiment is not a comparable large benchmark of model capabilities.

Gottal and Matthes [3] investigate verifiable BN parameterization from scientific literature by reconstructing probability information from empirical evidence. This differs from our input regime: our generator receives structured network descriptions and instructions, without a retrieval pipeline supplying external empirical sources. Fluent explanations and perceived internal knowledge do not supply the provenance that evidence-grounded parameterization seeks.

The references come from bnRep, a repository of BNs from academic literature [5]. The seismic reference originates in a probabilistic assessment of a reinforced-concrete building considering hazard and vulnerability [7]. The electric-vehicle reference concerns fire risk using a WBS–RBS and fuzzy BN approach [1]. They provide concrete targets with different state cardinalities and structures. Their CPTs are contextual reference judgments, not universal ground truth for every building or vehicle. Agreement with them is therefore reference agreement rather than empirical event calibration.

3 Reconstructed Study Design

3.1 Evidence and scope

We reconstructed the procedure from four generation notebooks, four generation ZIP archives, a V5 evaluation notebook, and four evaluation ZIP archives. Generation archives include network and persona specifications, manifests, raw responses, generated CPTs, and intermediate fitting records. Evaluation archives include reference exports, aligned comparisons, and aggregate and row-level metrics. Recorded generation dates span October 3–10, 2026; V5 evaluations are dated October 14. This reconstruction and the uniform baseline were produced retrospectively on October 16.

The notebooks and manifests establish executable settings and recorded operations. They do not establish why those settings were selected, whether they preceded inspection, or whether other attempts were discarded outside the supplied corpus. No original written protocol was available. Although the evaluator supports another network, `gasifier`, no corresponding experimental outputs were present, so it is excluded.

Table 1: Networks in the recovered corpus. Rows include roots; probabilities count all child states.

	Seismic	Electric vehicle
Nodes / edges	10 / 12	23 / 22
Root / conditional nodes	4 / 6	15 / 8
States per variable	3	2
Root / conditional rows	4 / 78	15 / 64
Total CPT probabilities	246	158

The design crosses two networks (Table 1), two sizes, and four prompt conditions. Each cell has one recorded full-network generation, labeled `r1`. The models are Qwen3.5-4B and Qwen3.5-9B from the Qwen namespace on Hugging Face. Conditions comprise a no-persona control and three personas. For seismic risk, P1 is a structural engineer, P2 a seismologist, and P3 a probabilistic risk analyst. For electric-vehicle risk, P1 is an electric-vehicle systems engineer, P2 an automotive fire-safety and reliability specialist, and P3 a probabilistic risk analyst. All conditions use the hybrid algorithm: "control" means absence of a persona, not absence of RN completion.

Each generator receives a fixed graph, variable descriptions, state names, and qualitative context. Reference CPTs are loaded separately by the evaluator; the generator does not call bnRep or consume those reference files. This separation limits direct leakage through the executed pipeline but cannot rule out familiarity with public material. A preliminary name-only familiarity question is archived, but self-report is not a contamination test.

3.2 Confidence-guided construction

Roots are elicited directly as FullDist probability vectors in the specified state order. For each eligible conditional node, the model first reports a score $c_r \in [0, 1]$ for every parent-configuration row r, together with a positive, negative, or uncertain direction for each parent. The prompt defines the score as confidence in possessing relevant internal knowledge for estimating the full distribution. It is not an event probability, an error bound, or a calibrated selective-prediction score.

Rows with $c_r \geq 0.80$ receive direct FullDist elicitation. Those distributions remain the final outputs for selected rows and also anchor RN fitting. Other rows are completed by RN. The threshold is recovered from the notebooks; its tuning history is unknown.

For a parent with K_j ordered states, state $s_j \in \{0, \ldots, K_j - 1\}$ is represented by midpoint $z_j = (s_j + 1/2)/K_j$. A negative direction maps this to $1 - z_j$; a positive direction retains it. Uncertain directions default to positive. With nonnegative weights w_j summing to one, the location parameter is $\mu = \sum_j w_j z_j$. Child-state probability is

$$q_k = \frac{\Phi((b_{k+1} - \mu)/\sigma) - \Phi((b_k - \mu)/\sigma)}{\Phi((1 - \mu)/\sigma) - \Phi(-\mu/\sigma)}, \quad b_k = k/K, \qquad (1)$$

where Φ is the standard-normal cumulative distribution and K is child cardinality. Completion thus imposes ordered, smooth distributions governed by a small parameter set. Ordering binary states as no<yes is an assumption; it does not establish that weighted means adequately represent failure gates or interactions.

With anchors, the code fits softmax parent weights and σ using confidence-weighted squared Hellinger loss plus 0.02 times squared distance from equal weights. L-BFGS-B uses up to 500 iterations, log-weight bounds $[-6, 6]$, and $\sigma \in [0.03, 0.45]$. Direction combinations are enumerated when anchor count is at least $\max(2, p+1)$ for $p \leq 6$ parents; otherwise direction hints are retained. Without anchors, the code uses equal weights and $\sigma = 0.16$, with the hints. We call this *unanchored RN*. It differs from a uniform-distribution error fallback. No archived rows are marked as uniform fallback.

A high-confidence answer therefore determines its own row and may influence other rows through fitted parameters. Low-confidence rows are constrained by RN, but constraint alone does not guarantee correctness. No-anchor completion still depends on model-supplied signs despite using default numeric parameters.

3.3 Execution and pooling

Manifests record 4-bit quantization, temperature 0.1, base seed 42, and disabled thinking. Per-call seeds derive from a hash and the base seed. Requested model revision is `main`, and the loaded model-commit field is null; exact weight snapshots are not recoverable from that field. The notebooks install development dependencies. Fixed seeds and archived code support inspection, but do not establish bitwise reproducibility across versions and devices.

For each network and model, two linear pools combine the three persona networks, excluding control. The equal pool assigns $1/3$ to each persona. The configured-weight pool uses $(0.5, 0.3, 0.2)$ in P1–P3 order. Seismic overrides are $(0.15, 0.70, 0.15)$ for `SeismicHazard` and $(0.70, 0.10, 0.20)$ for `Fragility`. For corresponding CPT rows,

$$q_{ir}^{\text{pool}}(k) = \sum_{a=1}^{3} \alpha_{ia} q_{ir}^{(a)}(k). \tag{2}$$

The artifacts do not establish performance-based learning or a prospective rationale for these weights. Pooling yields eight additional networks, for 24 outputs. These deterministic combinations add neither LLM calls nor independent repetitions.

3.4 Alignment and outcomes

The evaluator exports bnRep references and aligns predictions by node name, child-state name, and each parent name/state pair. Raw array position is insufficient: electric-vehicle exports require state and parent-order reconciliation. Both sizes share a reference-file SHA-256 hash within each domain. Name alignment prevents indexing errors but does not prove semantic equivalence between elicitation descriptions and source models.

We reproduce scalar mean absolute error (MAE) and root mean squared error (RMSE), and two equally weighted row metrics. For reference p_r and prediction q_r, total variation is $\text{TV}_r = \frac{1}{2}\sum_k |p_r(k) - q_r(k)|$. CPT KL is $d_r = D_{\text{KL}}(\widetilde{p}_r \| \widetilde{q}_r)$ in bits, where the tilde denotes normalization, addition of $\epsilon = 10^{-8}$ to every state, and renormalization. We report its arithmetic mean across all rows, including roots. Smoothing avoids infinite reported KL at zero predictions while retaining large penalties for confident disagreement.

The evaluator also reports a weighted network score,

$$\text{BN KL}_\epsilon = \sum_i \sum_r P_{\text{ref}}(\text{Pa}_i = r)\, d_{ir}, \tag{3}$$

with root weight one. Without smoothing, matching-graph joint KL admits this local decomposition. Here local KL is smoothed but parent weights come from the reference BN. Equation 3 is thus the evaluator's *smoothed weighted BN KL score*, not exact unsmoothed joint KL. Equal-row CPT KL emphasizes all specified configurations; BN KL emphasizes configurations probable under the reference. The latter sums over nodes, so its raw magnitudes are not a cross-domain difficulty ranking.

Our retrospective baseline assigns every child state probability $1/K$ on every row, evaluated with identical references, smoothing, and weighting. MAE depends on cardinality: with constant K, mean TV equals $K/2$ times scalar MAE. TV therefore equals MAE for the binary network and 1.5 times MAE for the ternary network. We emphasize TV and both KL summaries in the tables; all five metrics remain in the package.

Comparisons are descriptive. CPT rows share parents, prompts, fitted parameters, and networks; treating them as independent replicates would inflate the evidence. With one generation per cell, we do not report inferential significance, confidence intervals across runs, or population-level model rankings.

4 Results

4.1 RQ1: Domain and metric dependence

Seismic results (Table 2) do not give a consistent size ordering. The 9B model has lower TV, and hence MAE, in five of six matched conditions. However, its control has higher CPT KL (0.7929 versus 0.6997) and weighted BN KL (7.3764 versus 5.5247) than the 4B control. The lowest seismic BN KL is the 4B equal pool (3.5521), while the lowest TV is 4B P1 (0.2641). Several outputs fail to improve on uniform CPT KL (0.4560). Better absolute error does not imply better logarithmic agreement.

For electric-vehicle risk (Table 3), 4B has lower TV and CPT KL in all six matched conditions. It has lower BN KL in five, with P3 the exception. More importantly, *all twelve outputs have worse mean CPT KL than uniform probabilities*. The best observed CPT KL, 0.7895 for the 4B equal pool, is approximately 29.7% above uniform's 0.6087. This tempers conclusions drawn from a selected metric or only within-method comparisons.

Uniform does not dominate every outcome. The lowest electric-vehicle TV is 0.3667 for 4B P2, versus uniform's 0.4041, an improvement of approximately 9.3%. The 4B equal pool also improves BN KL over uniform (5.6100 versus 12.0939). These differences are compatible: a method can improve absolute distance or reference-weighted performance while making severe logarithmic errors on other configurations. Figure 1 shows the persistent electric-vehicle CPT KL deficit.

4.2 RQ2: Pooling helps selectively

For seismic 4B, equal pooling reduces BN KL by approximately 35.7% and CPT KL by 40.4% relative to control. Its BN KL improves on every constituent persona. Yet its TV (0.2770) remains higher than P1's (0.2641). Configured-weight pooling is worse than equal pooling on all three displayed seismic metrics for both sizes. Thus the specified weights offer no observed advantage over equal weighting in this domain.

Table 2: Seismic agreement. KL values are in bits; all columns are lower-is-better. BN KL denotes Equation 3. P1: structural engineer; P2: seismologist; P3: risk analyst. Uniform is model-independent and repeated for comparison.

	Qwen3.5-4B			Qwen3.5-9B		
Condition	BN KL	CPT KL	TV	BN KL	CPT KL	TV
Control	5.5247	0.6997	0.3346	7.3764	0.7929	0.2651
Persona 1	4.1567	0.4216	0.2641	8.0698	0.6877	0.2925
Persona 2	4.1447	0.5522	0.2959	4.9287	0.4342	0.2668
Persona 3	5.0014	0.6748	0.3216	4.1073	0.4263	0.2706
Equal pool	3.5521	0.4172	0.2770	4.2477	0.4241	0.2674
Weighted pool	3.7597	0.4622	0.2856	4.4610	0.4330	0.2707
Uniform	5.7130	0.4560	0.3069	5.7130	0.4560	0.3069

Table 3: Electric-vehicle agreement. Definitions match Table 2. P1: electric-vehicle systems engineer; P2: automotive fire-safety and reliability specialist; P3: risk analyst.

	Qwen3.5-4B			Qwen3.5-9B		
Condition	BN KL	CPT KL	TV	BN KL	CPT KL	TV
Control	9.1131	0.8839	0.4005	10.6434	1.6441	0.4170
Persona 1	17.8188	0.9941	0.3934	22.6972	2.1156	0.3992
Persona 2	11.4584	0.8578	0.3667	13.1395	1.5482	0.3971
Persona 3	31.3402	1.1653	0.3947	24.0429	2.9540	0.4277
Equal pool	5.6100	0.7895	0.3705	11.8444	0.9435	0.3986
Weighted pool	5.8970	0.7956	0.3739	11.5647	0.9370	0.3967
Uniform	12.0939	0.6087	0.4041	12.0939	0.6087	0.4041

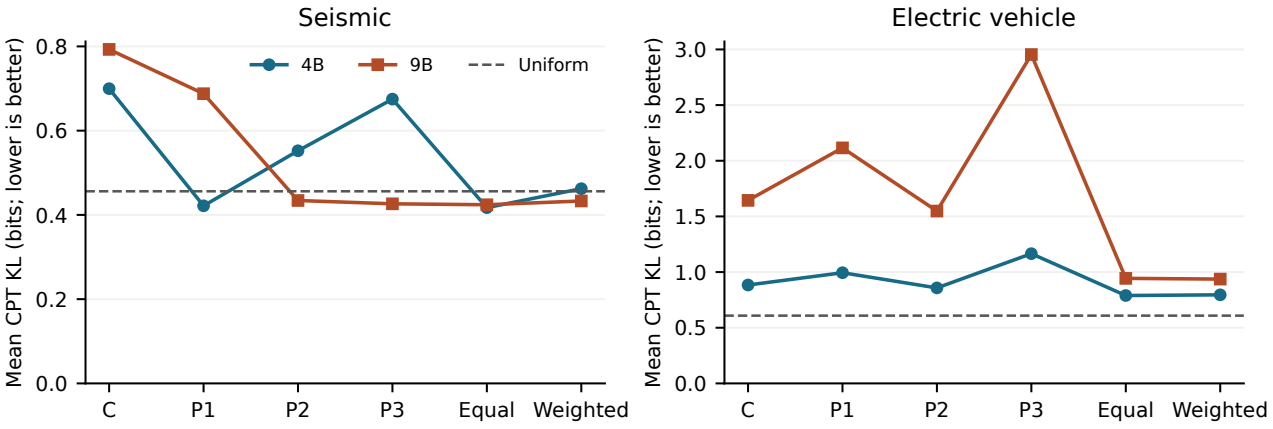


Figure 1: Mean CPT KL across conditions. C is the no-persona hybrid control; P1–P3 are personas. Lines connect categories for readability, not a continuous treatment. Uniform is computed separately for each domain.

For seismic 9B, the equal pool improves BN KL and CPT KL over control but slightly increases TV (0.2674 versus 0.2651). P3 retains lower BN KL than either pool. Aggregation changes the balance among errors rather than yielding one best output by every criterion.

For electric-vehicle 4B, equal pooling lowers BN KL from 9.1131 to 5.6100, a 38.4% reduction. Both pools improve control on all displayed outcomes, although P2 has lower TV than either. For 9B, pooling lowers CPT KL and TV but increases BN KL: 11.8444 for equal pooling versus 10.6434 for control, approximately 11.3% worse. Configured-weight pooling also has worse BN KL than control (11.5647).

Pooling can moderate incompatible or extreme predictions, but control is not a constituent and need not be improved by the combination. The results support reporting every constituent and both pools. They do not justify selecting weights using these reference scores and treating the improvement as an independent evaluation.

4.3 RQ3: Routing and concentrated failures

Most conditional rows use RN (Figure 2). Seismic 4B directly elicits only 3 of 78 conditional rows in control and zero in each persona. All six conditional nodes in each seismic 4B persona therefore use unanchored RN. Their conditional numeric tables arise from equal weights and default spread, influenced by qualitative signs, rather than fitting to directly elicited anchors.

Seismic 9B selects 12, 19, 12, and 10 conditional rows in control, P1, P2, and P3. Electric-vehicle 4B selects 0, 0, 2, and 0 of 64 conditional rows; 9B selects 4, 23, 18, and 17. The larger model uses direct elicitation more often in every matched condition. This describes routing behavior, not confidence calibration.

Directly elicited conditional rows expose a substantial electric-vehicle 9B failure mode. Their recorded confidences are 0.90–0.95. Their mean row KL values are 16.6158, 5.4595, 4.1100, and 10.5000 bits for control through P3. Although these groups contain only 4–23 rows, they account for approximately 51.2%, 75.1%, 60.5%, and 76.5% of the corresponding sums of unweighted row KL across complete networks. These percentages concern CPT KL, not weighted BN KL.

For example, the 9B electric-vehicle control assigns probability one to the yes state of `REI` when `CBI=no` and `HF=yes`; the reference assigns 0.0738. Smoothed row KL is approximately 24.23 bits. This is reference disagreement after a high-confidence routing decision. It does not establish the true event probability, but shows why the selection score is not an accuracy guarantee.

Large errors are not confined to screened rows. In electric-vehicle 4B P3, root `IS` receives probability one for yes against reference 0.105, producing approximately 23.30 bits of KL. Roots bypass the confidence gate and have unit weight in Equation 3. An audit restricted to conditional routing would miss this source of network-level error.

These observations cannot estimate the effect of FullDist versus RN: the methods process different, confidence-selected rows, and anchored RN may depend on FullDist outputs. A component comparison requires counterfactual outputs for the same rows under direct-only and RN-only conditions. This diagnosis identifies where a subsequent controlled experiment should focus.

5 Discussion

The hybrid procedure produces valid tables across the recovered cases, but validity does not imply satisfactory parameterization. The uniform comparison is instructive. A framework can appear useful through its best pooled weighted score while performing worse than a trivial distribution on mean row KL. Reporting both prevents favorable aggregation from concealing large local disagreements.

Three interacting mechanisms merit further study. First, confidence determines numerical flexibility: 9B takes the direct path more often, and some answers are extreme. Second, RN constrains many rows to a small distribution family, sometimes without numerical anchors. Third, pooling combines entire persona networks, affecting roots and conditional tables. Model-size and persona changes alter these behaviors together. None is independently established as the explanation for a score difference.

An unanchored RN table can serve as a transparent structural assumption, but its provenance should be visible. A practical interface should distinguish direct elicitation, anchored completion, and default-parameter completion. These categories carry different evidence even when a CPT editor displays equally precise decimals. Similarly, absence of an error fallback does not mean absence of assumptions: the no-anchor branch is a normal algorithmic path.

A stronger machine-learning experiment would fix descriptions and alignment, repeat generations across seeds, and compare direct-only, RN-only, and hybrid conditions on the same rows. It would vary the confidence threshold and assess retained-row error against coverage, selecting parameters on separate development networks. Persona weights and probability regularization should also be selected independently of final test references.

Downstream checks are necessary. CPT agreement does not directly measure whether posterior queries, rankings, or decisions improve. Query-level evaluation under representative evidence patterns could reveal whether local errors propagate to consequential outputs. Experts would then assess semantic fit and applicability of the source model. Current results motivate these tests rather than substitute for them.

6 Validity and Reproducibility

Internal validity. The protocol is reconstructed retrospectively. Parameter-selection timing, discarded trials, and original selection rationales are unavailable. Shared seeds are not independent repetitions. Personas change screening and fitting as well as prompt context, preventing single-component attribution. Direct-only and RN-only arms are absent.

Construct validity. Outcomes measure agreement with contextual benchmark parameters, not event calibration, expert quality, or operational safety. Self-confidence has no validated correctness mapping. Ordinal encoding and midpoint weighted means impose assumptions potentially unsuitable for some mechanisms. KL depends on the disclosed smoothing rule, particularly at zero probabilities.

External and statistical validity. Two networks and two sizes from one model family provide narrow coverage. Binary and ternary CPTs differ in size and error geometry; whole-network scores depend on node count and reference marginals. Each cell has one run, and pools are dependent derived outputs. We cannot quantify sampling variability or establish statistically reliable model superiority. Public-source memorization remains possible despite code-level separation from reference files.

Reproduction boundary. The package preserves eight original archives and five original notebooks with SHA-256 checksums and provenance. A dependency-free Python reanalysis reads generated CPTs, reconstructs persona pools, checks probability validity, aligns names, and verifies every predicted probability against evaluator exports. It recomputes five metrics for all 24 outputs within 10^{-7} tolerance, allowing small reference rounding differences, and derives uniform baselines. BN KL reuses archived reference-parent marginals after checking normalization; it does not independently repeat inference. The original evaluator and reference exports support that separate path.

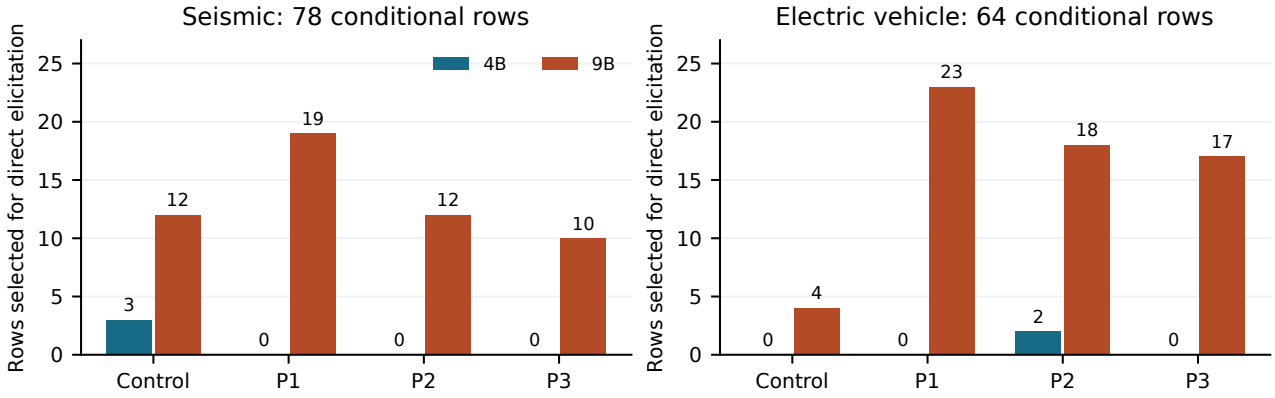


Figure 2: Conditional rows selected for direct FullDist elicitation. Other conditional rows use RN. Roots are always elicited directly and excluded here. Pools introduce no new routing decisions.

Fresh LLM generation is a different claim. Missing loaded model commits and development dependencies prevent a guarantee of identical outputs. The package documents these gaps rather than claiming a fully locked historical environment. Tables and figures derive from recomputed CSV files. The author repository is private during double-blind review; its identifying URL is omitted here. Original archives require anonymization before reviewer-facing distribution because notebook metadata and provenance may identify authors.

7 Conclusion

We reconstructed and reanalyzed confidence-guided LLM–Ranked Nodes parameterization on seismic and electric-vehicle risk networks. Sixteen generations and eight persona pools show substantial dependence on domain, condition, and metric. Equal pooling provides the lowest observed weighted BN KL in both domains with 4B, yet every electric-vehicle output underperforms uniform probabilities on mean CPT KL. More frequent direct elicitation by 9B includes highly confident, large disagreements, while many 4B conditional tables use unanchored RN defaults.

The evidence supports an exploratory account and reproducible basis for further study. It does not establish that confidence routing, personas, or RN completion causally improve accuracy. Evaluation should expose each row's source, include simple baselines, inspect extreme predictions, and connect local agreement to downstream queries before generated parameters enter an applied BN.

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